

Data Flow Diagram

Date	30 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through the Voice-Based Concept Understanding Analyzer. It represents the interaction between external entities, system processes, data stores, and the flow of information from voice input to report generation. The diagram provides a high-level overview of how audio data is processed, analyzed, evaluated, stored, and finally presented to the user.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded Shape	External Entity	A person, organization, or external system that sends or receives information from the Voice-Based Concept Understanding Analyzer.
Rectangle with Number	Process	Represents a processing activity such as Speech-to-Text, Semantic Analysis, Audio Feature Extraction, or Report Generation.

Rectangle (Data Store)	Data Store	Stores information such as uploaded audio files and generated reports for future retrieval.
Labeled Arrow	Data Flow	Indicates the movement of information between external entities, processes, and data stores.

Data Flow Diagram Visualization

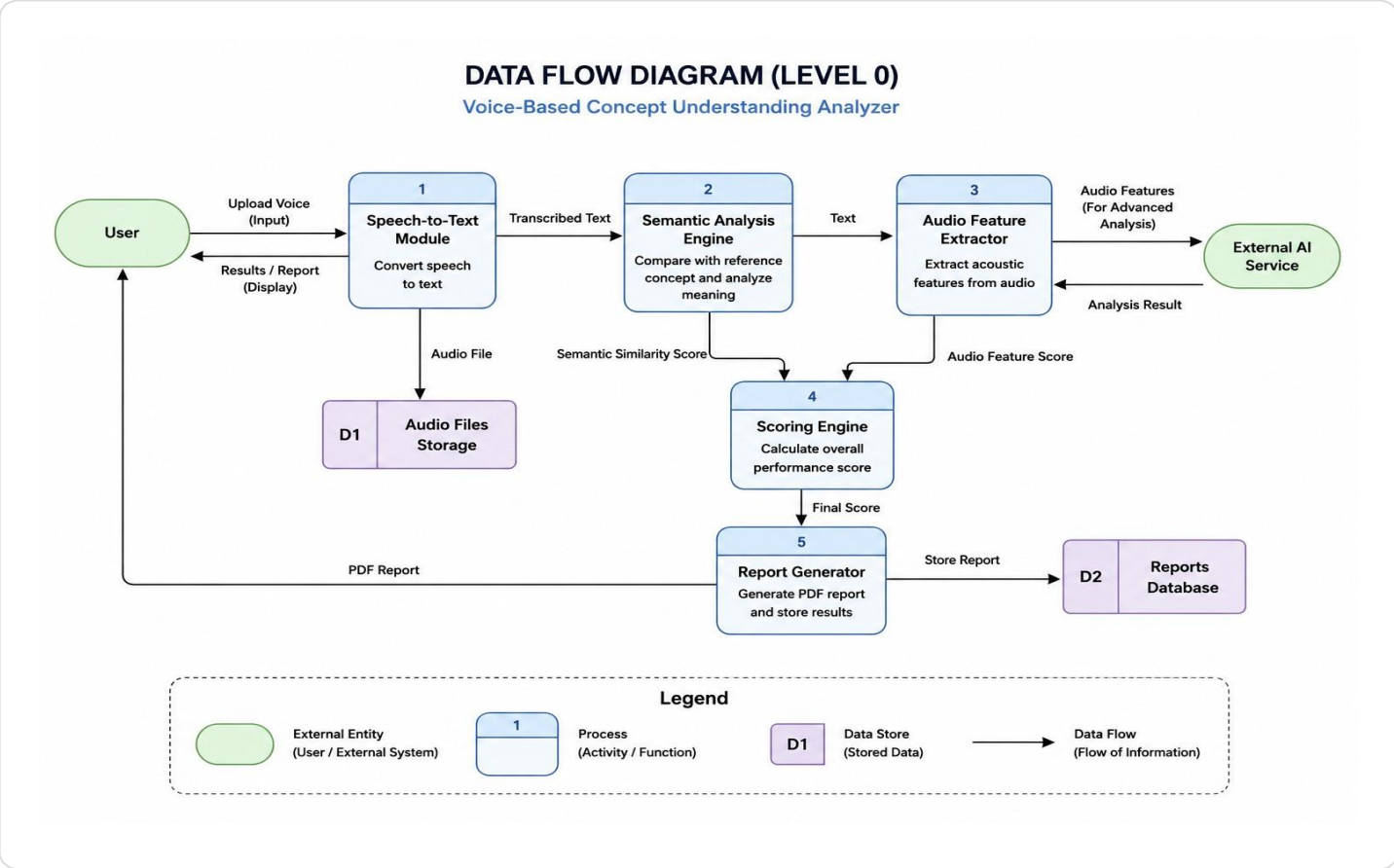


Figure 1: Data Flow Diagram of Voice-Based Concept Understanding Analyzer

Description

The workflow begins when the user uploads a voice recording through the application. The Speech-to-Text Module converts the audio into text, which is forwarded to the Semantic Analysis Engine to compare the user's explanation with reference concepts. Simultaneously, the Audio Feature Extractor analyzes speech characteristics such as pitch, clarity, and fluency. The Scoring Engine combines semantic similarity and audio feature scores to calculate the final

performance score. The Report Generator then creates a PDF report, stores it in the Reports Database, and displays the final evaluation results to the user.